Progressive exercises

E1 · Direct application. Solve in $[-\pi;\pi]$ the equation $\sin x = -\frac{1}{2}$.

E2 · Application and justification. Solve in $[-\pi;\pi]$ the inequality $\cos x \geq \frac{\sqrt{2}}{2}$.

E3 · Method to organise. We define on $\mathbb{R}$ the functions $f(x) = \cos(3x-1)$ and $g(x) = \sin^2 x$. Calculate their derivatives.

E4 · Choice of strategy. Consider $f(x) = 2 + \sin x$ on $[0;2\pi]$. Calculate its derivative and draw its table of variations.

E5 · Synthesis or proof. Calculate $\displaystyle\int_0^{\pi} \sin x \, \mathrm{d}x$, then the average value of the sine function on $[0;\pi]$.